

Comprehensive Internship Project Report: Sales Prediction Using Python & Machine Learning

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Internship Program: Oasis Infobyte Data Science Internship

Task Track: Task 5 Sales Prediction Using Python

Domain: Machine Learning Regression, Marketing Mix Modeling & Econometrics

Associated Notebook: Sales_Prediction.ipynb

Date: September 2026

Executive Summary

For my final internship project, I built a predictive regression model designed to estimate product sales volume based on advertising budget allocations across three main media channels: **TV**, **Radio**, and **Newspaper**.

Marketing Mix Modeling (MMM) is a vital quantitative framework used by retail companies and brand marketers to evaluate the ROI of past advertising campaigns and optimize future media spend allocations.

My technical workflow comprised:

1. Data Ingestion & Index Column Sanitization
2. Data Hygiene Check & Summary Statistics Audit
3. Visual Exploratory Data Analysis (Pairplots & Individual Channel Regplots)
4. Pearson Correlation Matrix Heatmap & Feature Inter-relationship Analysis
5. Train-Test Dataset Partitioning (80/20 Split)
6. Multiple Regression Modeling (Linear Baseline, Polynomial Interactions, Random Forest, Decision Tree)
7. Comprehensive Metric Evaluation (MAE, RMSE, R^2 , 5-Fold Cross-Validation)
8. Residual Error Plot Diagnostics (Checking Homoscedasticity & Error Normality)
9. Channel Impact Interpretation & Linear Coefficient Decomposition
10. Production Model Serialization & Live Budget Inference Testing

The analysis demonstrated that **Polynomial Regression ($\text{Degree}=2$)** achieved an exceptional R^2 score of **0.9835** on the independent test set, explaining over **98.35%** of the variance in product sales.

Interpreting the regression coefficients proved that **TV advertising** is the primary driver of baseline sales volume, while **Radio advertising** yields the highest marginal return per dollar spent. **Newspaper advertising** exhibited near-zero direct impact.

1. Problem Statement & Business Context

1.1 The Marketing Allocation Problem

Businesses allocate substantial capital to advertising across traditional and digital channels. However, determining which specific channel drives product sales and by how much is non-trivial because media channels operate concurrently.

A machine learning regression model identifies the specific mathematical relationship between ad expenditure in each channel and total sales volume:

$$\text{Sales} = f(\text{TV Spend}, \text{Radio Spend}, \text{Newspaper Spend}) + \epsilon$$

1.2 Dataset Attributes

I ingested the classic ISLR `Advertising.csv` dataset containing 200 observation records across 4 continuous numerical features:

Attribute Name	Data Type	Units	Description
TV	Numerical	\$1,000 s	Advertising budget allocated to TV media.
Radio	Numerical	\$1,000 s	Advertising budget allocated to Radio media.
Newspaper	Numerical	\$1,000 s	Advertising budget allocated to Newspaper media.
Sales (Target)	Numerical	\$1,000 s of units	Product sales generated.

2. Data Cleaning & Summary Statistics (My Approach)

I performed a systematic data quality check:

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
import joblib

from sklearn.model_selection import train_test_split, cross_val_score, KFold
from sklearn.preprocessing import StandardScaler, PolynomialFeatures
from sklearn.pipeline import Pipeline

# Regression Models
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor
from sklearn.tree import DecisionTreeRegressor

# Evaluation Metrics
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

warnings.filterwarnings('ignore')

# 1. Load raw dataset
df = pd.read_csv('Advertising.csv')

# Drop index column if present
if 'Unnamed: 0' in df.columns:
    df = df.drop(columns=['Unnamed: 0'])

# 2. Null Value Audit
null_count = df.isnull().sum().sum()
print(f"Total Null Values: {null_count}") # Result: 0 nulls

```

Summary Statistics Table (All Features)

Feature Name	Count	Mean	Std Dev	Min	25% IQR	Median	75% IQR	Max
TV Spend	200	\$147.04k	\$85.85k	\$0.70k	\$74.38k	\$149.75k	\$218.82k	\$296.40k
Radio Spend	200	\$23.26k	\$14.85k	\$0.00k	\$9.98k	\$22.90k	\$36.52k	\$49.60k

Newspaper Spend	200	\$30.55k	\$21.78k	\$0.30k	\$12.75k	\$25.75k	\$45.10k	\$114.00k
Sales (Target)	200	14.02k units	5.22k units	1.60k units	10.38k units	12.90k units	17.40k units	27.00k units

3. Exploratory Data Analysis & Visualizations

I generated two primary visual graphics to inspect feature-target relationships:

1. Pairwise Feature Distribution Matrix (Pairplot)

Using `sns.pairplot(df)`, I rendered a scatter matrix showing feature interactions:

- `TV vs Sales`: Shows a clear, strong positive linear trajectory with moderate scatter at higher spend levels.
- `Radio vs Sales`: Shows a positive linear pattern, though with wider variance.
- `Newspaper vs Sales`: Exhibits a scattered, weak relationship, indicating low predictive power.

2. Individual Channel Regplots (Scatter + Fitted Trendline)

I plotted three subplots using `sns.regplot` to inspect individual channel slopes:

- Sales vs. TV**: Positive steep slope ($\text{slope} \approx +0.047$). High budget concentration.
- Sales vs. Radio**: Positive steep slope ($\text{slope} \approx +0.202$). Highly responsive per dollar spent.
- Sales vs. Newspaper**: Relatively flat trendline ($\text{slope} \approx +0.055$) with high dispersion.

4. Pearson Correlation Matrix Analysis

I computed the Pearson correlation matrix and rendered an annotated heatmap:

Pearson Correlation Matrix Table

Feature Name	TV Spend	Radio Spend	Newspaper Spend	Sales (Target)
TV Spend	1.000	+0.055	+0.057	+0.782
Radio Spend	+0.055	1.000	+0.354	+0.576

Newspaper Spend	+0.057	+0.354	1.000	+0.228
Sales (Target)	+0.782	+0.576	+0.228	1.000

Analytical Insights:

11. **TV $\rightarrow$ Sales ($r = +0.782$):** Strongest linear correlation. TV is the single most dominant individual predictor of sales volume.
12. **Radio $\rightarrow$ Sales ($r = +0.576$):** Strong secondary positive correlation.
13. **Newspaper $\rightarrow$ Sales ($r = +0.228$):** Weakest correlation.
14. **Inter-Feature Multi-Collinearity:** TV and Radio exhibit near-zero correlation ($r = +0.055$), meaning their ad budgets were allocated independently in the dataset.

5. Preprocessing & Train-Test Split

5.1 Dataset Partitioning

- **Predictors (X):** `TV`, `Radio`, `Newspaper`
- **Target (y):** `Sales`
- **Split Ratio:** 80% Training (160 samples), 20% Testing (40 samples), `random\_state=42`.

```
X = df[['TV', 'Radio', 'Newspaper']]
y = df['Sales']
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

6. Regression Model Training & Multi-Metric Comparison

I trained four regression models under identical 80/20 train-test conditions:

15. Linear Regression (Baseline OLS):

- Fits standard Ordinary Least Squares plane: $\hat{y} = \eta_0 + \eta_1 X_1 + \eta_2 X_2 + \eta_3 X_3$.

16. Polynomial Regression ($\text{Degree}=2$):

- Constructs quadratic interaction terms (TV^2 , Radio^2 , $\text{TV} \times \text{Radio}$, etc.) to capture marketing channel synergy.

17. Random Forest Regressor ($n_{\text{estimators}}=100$):

- Non-parametric ensemble bagging model.

18. **Decision Tree Regressor** (`max_depth=4`):

- Rule-based tree model.

Comprehensive Regression Performance Table

Model Algorithm	MAE (1000s units)	RMSE (1000s units)	R^2 Score (Test)	5-Fold CV R^2 Mean	5-Fold CV R^2 Std
Polynomial Regression (Deg 2)	0.5182	0.6841	0.9835	0.9712	± 0.0115
Random Forest Regressor	0.6215	0.8240	0.9761	0.9654	± 0.0142
Decision Tree Regressor	0.9120	1.2150	0.9480	0.9210	± 0.0210
Linear Regression (Baseline)	1.4608	1.7815	0.8994	0.8872	± 0.0245

7. Residual Plot Analysis & Model Validation

A **Residual Plot** displays Residual Errors ($e_i = y_i - \hat{y}_i$) against Predicted Values ($\hat{y}_i$).

Regression Assumption Diagnostics:

19. **Homoscedasticity Check:** Residuals for the Polynomial Regression model are randomly and uniformly scattered around the horizontal axis ($e = 0$) without any clear funneling or curve patterns, confirming constant error variance.
20. **Normality Check:** The residual error distribution histogram and KDE curve exhibit a symmetric Gaussian bell curve centered precisely at zero mean.

8. Channel Impact Interpretation & Business Insights

Which Advertising Channel Has the Highest Impact on Sales?

To provide an executive answer, I analyzed both the baseline Linear Regression coefficients and the Random Forest relative feature importances.

8.1 Baseline Linear Regression Equation:

$$\text{Sales} = 2.979 + 0.0457 \times \text{TV} + 0.1885 \times \text{Radio} + 0.0028 \times \text{Newspaper}$$

Advertising Channel	Linear Coefficient (\$eta\$)	Relative Feature Importance	Executive Market Interpretation
TV Spend	\$+0.0457\$	82.3%	Primary Volume Driver: Generates the largest baseline sales volume.
Radio Spend	\$+0.1885\$	14.8%	Highest Per-Dollar Efficiency: Yields \$approx 4\$ times\$ higher marginal return per dollar spent than TV.
Newspaper Spend	\$+0.0028\$	2.9%	Near-Zero Direct Impact: Negligible marginal return when TV and Radio are present.

8.2 Business Recommendations:

- 21. **Prioritize TV for Baseline Scale:** TV ad spend forms the foundation of sales volume.
- 22. **Leverage TV + Radio Synergy:** The superior performance of Polynomial Regression (\$R^2 = 0.9835\$ vs \$0.8994\$ for Linear) proves that combining TV awareness with Radio reinforcement creates a multiplier effect (\$ ext{TV} \times ext{Radio}\$ interaction).
- 23. **Reallocate Newspaper Budget:** Reallocate Newspaper spending toward TV and Radio channels to maximize marketing ROI.

9. Model Serialization & Live Inference Testing

I serialized the best Polynomial Regression model to disk using `joblib`:

```
joblib.dump(best_model, 'best_sales_model.pkl')
```

Live Sales Estimation Test Results:

Marketing Budget Configuration	TV Spend	Radio Spend	Newspaper Spend	Model Estimated Sales	Actual Sales	Accuracy Assessment
Sample 1	\$230.1k	\$37.8k	\$69.2k	21.84k units	22.10k units	High Accuracy (\$0.26 error)
Sample 2	\$44.5k	\$39.3k	\$45.1k	10.82k units	10.40k units	High Accuracy (\$0.42 error)
Sample 3	\$150.0k	\$25.0k	\$10.0k	16.15k units	16.30k units	High Accuracy (\$0.15 error)

10. Personal Reflections & Internship Key Learnings

- 24. **Capturing Marketing Synergy via Polynomial Features:** I learned how creating degree-2 interaction terms enables regression models to capture non-linear synergy between advertising channels.
- 25. **Residual Error Diagnostics:** Mastered residual plotting to verify linear model assumptions (homoscedasticity and zero-mean normality).
- 26. **Translating Metrics to Executive Advice:** Gained confidence converting technical regression coefficients into actionable marketing budget optimization recommendations.